Suggested problems: 1-5

Problem 1: Suppose that $X \sim \mathcal{U}nif(a, b)$ is a **discrete** uniform random variable. Compute the expectation $E(X)$.

Problem 2: Suppose $X \sim \mathcal{B}er(p)$. Compute the expectation $E(X)$.

Problem 3: (Example 4.1.2) A stock purchased for \$18 is worth $18 + X$ a year later, where X is a discrete random variable with p.f. given by

$$f(x) = \begin{cases} 0.1 & : x = -2, \\ 0.4 & : x = 0, \\ 0.3 & : x = 1, \\ 0.2 & : x = 4, \\ 0 & : \text{else.} \end{cases}$$

(a) Compute the expected value $E(18 + X)$ of the stock a year later.

(b) Suppose that a person is deciding between investing their \$18 in the stock, or a (risk-free) savings account earning 5% annual interest. Which of the two investments would they choose if they were risk averse? Risk seeking? Risk neutral?

Problem 4: Suppose that $X \sim \mathcal{U}nif(a, b)$ is a **continuous** uniform random variable. Compute the expectation $E(X)$.

Problem 5: (Example 4.1.6) An appliance has a maximum lifetime of one year. The time X until it fails is a random variable with p.d.f. given by

$$f(x) = \begin{cases} 2x & : 0 < x < 1, \\ 0 & : \text{else.} \end{cases}$$

Compute the average time till failure.

Problem 6: (Example 4.1.11) Suppose that X is a continuous random variable with p.d.f. given by

$$f(x) = \begin{cases} 3x^2 & : 0 < x < 1, \\ 0 & : \text{else.} \end{cases}$$

Compute the expectation of $Y = 1/X$.